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Project 2 Report

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Abstract

This project explores three data mining tasks on healthcare datasets: association-rule mining, sequential pattern discovery, and predictive modeling. The Apriori algorithm is applied to a symptom–disease dataset to extract frequent symptom item sets and interpretable association rules. The Breast Cancer Wisconsin dataset is discretized into abnormality sequences and analyzed with a custom GSP algorithm, revealing malignant- and benign-specific feature progressions. Finally, a web application integrates supervised disease prediction, Apriori-based symptom suggestions, cancer-risk classification, and pattern-based interpretability, demonstrating how classical pattern mining and modern machine learning can jointly enhance clinical decision support.

Task 1- Apriori algorithm

The first part of this task consisted of data cleaning. In the original set, there were some duplicates and null, which we eliminated.

```
df = df.drop_duplicates()
```

Apart from getting rid of null values, another thing to take into account was the synonyms. There were 2 cases of similar Symptoms bearing the same name, so they were converted to a single one

```
# consider synonyms
synonyms = {'pyrexia': 'fever', 'skin_rash': 'rash'}
for col in symptom_num:
    df[col] = df[col].replace(synonyms)
```

Then, in order to ensure a flat group, it was ensured that all transactions follow the same normalized pattern (lowercase, no spaces, etc)

```
symptom_num = [f'Symptom_{i}' for i in range(1, 18)]
for col in symptom_num:
    df[col] = df[col].str.lower().str.strip()
```

An important aspect to ensure was a specific threshold for our data. In case there are too few transactions, initially, the dataset is extended in order to create more reference data.

```
    min_transactions = 1000
if len(transactions) < min_transactions:
    transactions = transactions * (min_transactions // len(transactions) + 1)
```

The Apriori algorithm works in a straightforward manner explained below. In the first part (finding popular itemsets) there are 3 main steps: compute, prune, build bigger.

The initial case of 1 item datasets can be taken into consideration for simplicity. In the **computation step**, all symptoms are counted in order to determine their support. The algorithm iterates through all transactions, finding the count of each symptom. After that, the count is divided by the total number of transactions:

```
    for i, count in all_items.items():
      support= count/num_tx
      item_supp[i]=support
```

When it comes to the **pruning step**, based on the results obtained previously, only the items that meet the minimum threshold can continue, discarding the ones below that:

```
        support=count/len(transactions)
        if support>= min_support:
          freq_items_Lnext[i]= support
```

In our algorithmic logic however, this step is inside the third part, **the iterative logic ('build bigger itemsets')**.

The main idea of apriori is that for each frequent set, its subsets must also be frequent, and for an infrequent sets then all its supersets can not be frequent either. Following this rational, the algorithm tries to iteratively grow the supersets until there is no increase in frequency anymore.

The number of items in each set is increased in each iteration. This is accomplished by combining the previous items in the previous set in order to create a larger one:

```
L1 = { {A}, {B}, {C} } → combine them →
candidates for L2 = { {A,B}, {A,C}, {B,C} }
```

After that, step 1 and 2 are repeated by counting the support and pruning each set that does not comply to the support threshold. Once this is completed, we continue with the next set until there are no candidates anymore;

```
L2 = { {A,B}, {B,C} } → merge → {A,B,C}
```

The second part consists of rules generation. The key principle here is crating rules for itemsets that have *if-then* relationships. For each frequent itemset discovered previously, we calculate its confidence given another one. A formula for confidence is:

$$\text{confidence}(A \rightarrow B) = \frac{\text{support}(A \cup B)}{\text{support}(A)}$$

```
for prev in itertools.chain.from_iterable(
    itertools.combinations(itemset, r) for r in range(1, len(itemset))
):
   [...]
        if conf >= min_confidence:
            rules.append((prev, consequent, supp, conf))
```

The algorithm can be run in the following way, resulting in 42 rules:

```
result = apriori(0.1, transactions)
rules = make_rules(result, min_confidence=0.7)
```

Task 2 – Mining Cancer Feature Patterns

2.1 Dataset and Preprocessing

We use the `Breast Cancer Wisconsin (Diagnostic)` dataset (569 patients, 30 numerical features). Each patient is labelled as `Malignant (M)` or `Benign (B)`.

Because sequential pattern mining requires categorical sequences, we turned each patient's feature vector into an ordered list of itemsets. The pipeline is summarised below:

1. **Quantile discretisation (3 bins).** Each feature was discretised into *low*, *medium*, *high*.

   ```
   disc = KBinsDiscretizer(n_bins=3, encode="ordinal", strategy="quantile")
   disc_data = disc.fit_transform(df[features])
   ```

2. **Z-score anomaly ranking.** For each patient we computed absolute z-scores to identify unusually large or small features:

```
z = (df[features] - df[features].mean()) / df[features].std()
abs_z = z.abs()
```

3. **Top-K extreme features.** For each patient, the $K = 10$ features with the highest absolute z-scores were selected.

4. **Tie grouping into itemsets.** Features with identical z-score ranks were grouped together in the same itemset:

$$\langle \{f_1\}, \{f_2, f_3\}, \{f_4\}, \dots \rangle.$$

5. **Split by class.** This produced two sequence databases: **212 malignant** sequences and **357 benign** sequences.

This transformed representation captures morphological abnormality as an ordered progression of discretised features.

2.2 GSP Algorithm

We implemented GSP from scratch. The algorithm iteratively:

1. generates frequent 1-sequences,
2. forms candidate k-sequences from frequent $(k-1)$-sequences,
3. counts supports with subsequence matching,
4. prunes candidates below the minimum support threshold.

The core support-counting routine:

```
def count_support(sequences, candidates):
    counts = {c: 0 for c in candidates}
    for seq in sequences:
        for cand in candidates:
            if is_subsequence(cand, seq):
                counts[cand] += 1
    return counts
```

2.3 Class-Conditional Support

Since class sizes differ (212 malignant vs. 357 benign), supports were normalised as:

$$\text{support}_M(s) = \frac{\#\{\text{malignant patients containing } s\}}{212}, \qquad \text{support}_B(s) = \frac{\#\{\text{benign patients containing } s\}}{357}.$$

This allows a meaningful comparison of patterns between classes.

2.4 Malignant-Specific Patterns (Ranked by Support)

Running GSP produced many frequent patterns. Below we present a curated selection of the **most relevant malignant-specific patterns**:

- the **most frequent length-2** malignant patterns, and
- the **most frequent length-3** malignant patterns,

all of which satisfy $\text{support}_B(s) = 0.00$ (never occur in benign cases) and are shown **ranked by descending malignant support**.

Sequential Pattern	**Support**$_M$	**Support**$_B$
$\langle\{\text{high_area_mean}\}, \{\text{high_perimeter_mean}\}\rangle$	**0.23**	0.00
$\langle\{\text{high_radius_worst}\}, \{\text{high_perimeter_mean}\}\rangle$	0.20	0.00
$\langle\{\text{high_area_worst}\}, \{\text{high_perimeter_mean}\}\rangle$	0.18	0.00
$\langle\{\text{high_area_mean}\}, \{\text{high_radius_mean}\}, \{\text{high_perimeter_mean}\}\rangle$	0.11	0.00
$\langle\{\text{high_area_worst}\}, \{\text{high_radius_mean}\}, \{\text{high_perimeter_mean}\}\rangle$	0.11	0.00

Interpretation

All listed sequences are strongly associated with malignant tumours and absent in benign cases. Several observations arise:

- The highest-support malignant patterns consistently combine **high area**, **high radius**, and **high perimeter**, reflecting the following tumour characteristics:
  - **Large tumours** (high mean/maximum area and radius),
  - **Rapid growth** (worst-case radius and area),
  - **Irregular borders** (high perimeter measures).
- The most frequent malignant patterns include both length-2 and length-3 sequences:

  $$\langle\{\text{high_area_mean}\}, \{\text{high_perimeter_mean}\}\rangle \quad (\text{support}_M = 0.23)$$

  and the length-3 sequences

  $\langle\{\text{high_area_mean}\}, \{\text{high_radius_mean}\}, \{\text{high_perimeter_mean}\}\rangle$, $\langle\{\text{high_area_worst}\}, \{\text{high_radius_mea}$
- The fact that $\text{support}_B = 0.00$ for all these sequences indicates that benign tumours never show these combinations of high morphological irregularity.
- These patterns align strongly with known morphological hallmarks of malignancy, validating the clinical interpretability of the GSP mining process.

2.5 Benign-Specific Patterns

Similarly, GSP identified several benign-specific sequences. Below we report the most relevant patterns (highest supports, lengths 2–3), all with $\text{support}_M(s) = 0.00$.

Sequential Pattern	**Support**$_B$	**Support**$_M$
$\langle\{\text{high_radius_mean}\}, \{\text{high_perimeter_worst}\}\rangle$	**0.14**	0.00
$\langle\{\text{high_perimeter_mean}\}, \{\text{high_perimeter_worst}\}\rangle$	0.13	0.00
$\langle\{\text{high_perimeter_mean}\}, \{\text{high_radius_worst}\}\rangle$	0.13	0.00
$\langle\{\text{high_radius_mean}\}, \{\text{high_perimeter_mean}\}, \{\text{high_radius_worst}\}\rangle$	0.10	0.00
$\langle\{\text{high_radius_mean}\}, \{\text{high_perimeter_mean}\}, \{\text{high_perimeter_worst}\}\rangle$	0.10	0.00

Interpretation

Benign sequences show a different structure:

- They involve **high radius** and **high perimeter**, but not high area (unlike malignant patterns).
- Supports range from 10% to 14% among benign tumours.
- They never appear in malignant cases.

This suggests some benign tumours may be large, but lack the irregular, expansive growth characteristic of malignancy.

2.6 Sensitivity Check Across Binning Strategies

To assess how the choice of discretisation affects the extracted sequential patterns, we compared three binning strategies available in `KBinsDiscretizer`:

- **Uniform:** Divides the feature range into equally sized intervals.
- **Quantile:** Divides the feature distribution so that each bin contains approximately the same number of samples.
- **K-Means:** Clusters feature values into k bins using 1D k-means.

When using **uniform** or **k-means** binning, the vast majority of features in the discovered patterns fell into the *medium* bin. This is likely not desired, because it makes it difficult to interpret which features are truly extreme or abnormal. In other words, almost all sequential patterns contain only "medium" items, reducing the practical insight that can be drawn from them.

By contrast, **quantile discretisation** produces bins that reflect the actual distribution of each feature. High values (e.g., top percentiles) are consistently

assigned to the *high* bin, allowing clear interpretation of patterns such as "high area", "high radius", or "high perimeter" in the extracted sequences.

This demonstrates that the choice of binning strategy strongly impacts the interpretability of sequential patterns, and for this dataset, quantile discretisation is preferable.

2.7 Discussion

The GSP mining results reveal clear differences between malignant and benign tumours:

- **Malignant tumours** follow patterns with **high area**, **high radius**, and **high perimeter**, often including worst-case measures. These patterns indicate **large tumours with rapid growth and irregular boundaries**, consistent with known hallmarks of malignancy.
- **Benign tumours** may have high radius or perimeter, but the patterns do not include high area or extreme combinations. This suggests **large but more uniform tumours**, without the irregular growth seen in malignant cases.
- Comparing supports between classes highlights **discriminative patterns**, features that clearly separate malignant and benign cases.
- Using top-K feature sequences turns raw measurements into **interpretable patterns**, showing how tumour features progress differently in each class.
- **Limitations:**
  - The choice of K and number of bins affects the patterns—different values could give slightly different results.
  - GSP may miss patterns that occur rarely but are still clinically relevant.
  - The method captures sequential order of features but not complex interactions between multiple features simultaneously.

Overall, GSP on discretised feature sequences provides a useful and interpretable way to study tumour morphology, but results should be considered alongside other statistical analyses and clinical knowledge. .

Task 3- Open advanced task

3.1 Application Description

The application idea was conceived on the notion of potential effectiveness and ease of use for consumers, particularly those in frequent contact with the aforementioned datasets. Therefore, a web-based medical symptom checker was developed for disease prediction and cancer risk assessment.

Primarily, both the disease-symptom and Wisconsin Breast Cancer datasets are used to predict likely diseases from symptoms and assess cancer risk from tumor features, providing interpretable results with Flask API and React UI.

3.2 Solution Overview

For disease prediction, a supervised classifier learns mappings from binary symptom vectors to diseases, while Apriori association-rule mining discovers frequently co-occurring symptoms to: 1) provide intelligent "related symptoms" suggestions and 2) calibrate the model's confidence when user inputs are partial or noisy. For cancer risk assessment, a Gradient Boosting model predicts malignant vs benign, augmented by a sequential pattern detector (GSP-style) that highlights discriminative feature progressions for interpretability. The UI supports manual input and CSV upload, shows confidence breakdowns and top candidates, and surfaces pattern evidence (rules for symptoms; sequential patterns for cancer features). Experimentally, the cancer model achieves a test accuracy of 92.98%, the disease model performs perfectly on the small benchmark dataset, Apriori mining yields 1,456 association rules in 41 diseases, and the sequential pattern module surfaces 30+ concise cancer-related feature patterns—together providing both accuracy and human-readable explanations.

Disease prediction begins by transforming each encounter into a binary vector $\mathbf{x} \in \{0,1\}^S$ over the curated symptom vocabulary ($S \approx 131$). A Random Forest classifier (`n_estimators=100`, `max_depth=20`, `min_samples_split=5`, `random_state=42`) is trained with stratified splits; inference produces class probabilities $p(d \mid \mathbf{x})$ (observed test accuracy 100% on the 304-case dataset).

In parallel, Apriori mining runs offline (`min_support=0.05`, `min_confidence=0.6`) to extract 1,456 frequent itemsets and association rules plus disease-specific frequent symptom sets (patterns found for all 41 diseases). At prediction time, a disease pattern-match score $m(d \mid \mathbf{x})$ is computed by comparing selected symptoms to each disease's top frequent sets (Jaccard similarity weighted by pattern support), then converted to a percentage $a = 100 \cdot m(d \mid \mathbf{x})$.

Scores are combined with a conservative, monotonic rule: if $a >$ base confidence b, use a boost $(0.6b + 0.4a)$; otherwise keep the strong base signal and add a small validation bonus $(b + 0.1a)$, capped at 100. This prevents unwarranted drops when rules are weaker while still rewarding strong matches. The top–5 diseases are returned with enhanced confidence plus rule-derived symptom suggestions from association consequents (e.g., the GERD set yields additional related "ulcers_on_tongue" when missing).

Cancer risk assessment takes the 30 diagnostic features, removes non-predictive IDs/unnamed columns, imputes when needed, and scales via `StandardScaler`. A Gradient Boosting classifier (`n_estimators=100`, `learning_rate=0.1`, `max_depth=5`, `random_state=42`) predicts malignancy (test accuracy 92.98%; key feature: `perimeter_worst` dominating importance). For pattern-level explanations, features are discretized using `KBinsDiscretizer` (quantile binning into low/med/high), and concise ordered patterns (GSP-style) are mined that differentiate malignant vs. benign cohorts (32 high-support discriminative sequences).

At inference, the user's input is discretized and matched against the mined patterns to surface the strongest aligned motifs alongside the model's probability, reinforcing interpretability without altering core decision boundaries.

3.3 Results

3.3.1 Disease Prediction

GERD for example, do not require a perfect or uniform set of symptoms for diagnosis. Its presentation vary widely across individuals, and symptoms often overlap or appear in only partial forms. These conditions can also coexist, making diagnosis more complex. Such variability highlights the value of pattern-mining methods like the Apriori algorithm for identifying meaningful symptom associations.

Symptoms	**Probability**
acidity;ulcers_on_tongue;vomiting;cough;chest_pain	74.53
stomach_pain;acidity;ulcers_on_tongue;vomiting;chest_pain	76.63
stomach_pain;acidity;ulcers_on_tongue;cough;chest_pain	76.84
stomach_pain;acidity;vomiting;cough;chest_pain	60.20
stomach_pain;ulcers_on_tongue;vomiting;cough;chest_pain	63.77
stomach_pain;acidity;ulcers_on_tongue;vomiting;cough;chest_pain	85.32

Table 1: GERD symptom combinations and corresponding predictions

The monotonic increase with richer, high-support GERD symptom clusters and the large jump only when the canonical full set is present empirically support that Apriori is functioning as a pattern-informed confidence calibrator rather than a random booster. This reinforces clinically specific combinations strongly, while leaving weaker sets at appropriately lower scores.

3.3.2 Cancer Diagnosis

As mentioned in Section 2.7, GSP mining revealed malignant cells as those with high area, radius, and perimeter - while benign cases may show high radius or perimeter alone, but lack high area and extreme combinations.

A specific malignant case was tested, and the top detected malignant patterns were analyzed.

- high area mean $\rightarrow$ high perimeter mean (0.230)
- high radius worst $\rightarrow$ high perimeter mean (0.200)
- high area worst $\rightarrow$ high perimeter mean (0.180)

If a tumor has a large area or radius, it often also has a large perimeter, and this is strongly associated with malignant tumors. These higher values reflect rapid, irregular, and aggressive tumor growth, typical of malignancy.